

ALEX MADARAS

JUNIOR BACK-END ENGINEER

ABOUT ME

I'm a 24-year-old Computer Science Master's graduate and aspiring Software Engineer with a passion for building meaningful software, particularly mobile and full-stack web applications that enhance people everyday's life, some of which I've showcased in my [web portfolio](#). I'm driven by curiosity and enjoy experimenting with new technologies, often integrating them into both personal and professional projects. I thrive in collaborative teams where innovation, learning, and shared motivation are valued.

EDUCATION

University of Twente

Master's in Computer Science

Courses: Software Engineering Techniques, Software Management, Requirement Elicitation, Software Architecture, Microservices Architecture in Web Development, Machine Learning, Exploratory Data Analysis.

Enschede, The Netherlands

Graduated

March 2025

West University of Timisoara

Bachelor's in Computer Science

Courses: Algorithms and Data Structures, Introduction to Python, Software Engineering, Practice Stage, Statistics and Numerical Analysis, Web Technologies, Artificial Intelligence, Object-Oriented Programming

Timisoara, Romania

2019-2022

IADE Creative University

Erasmus+ Semester - Bachelor of Game and Apps Development

Courses: Collaborative Learning, Mobile Programming in Kotlin, Game Frameworks, Agile Management for Multiplatform Projects.

Focus: building a Multiplatform Project consisting of a Windows 2D Game in Unity connected to a Companion App in Android through an SQL Database.

Lisbon, Portugal

2020-2021

WORK EXPERIENCE

ING Nederland B.V.

AI Engineer Intern

Amsterdam, Netherlands

Apr 2024 - Nov 2024

- Researched AI algorithms to detect data quality issues in credit risk data as part of my Master's thesis, achieving a **grade of 8.0/10**.
- Conducted top exploratory data analysis practices before the Model Development phase using Python, which resulted in detecting and removing **17% of irrelevant outliers (confirmed by the experts)**, which enhanced the accuracy of the anomaly detection process.
- Built, trained, and validated a Deep Learning model (Autoencoders) using Keras, detecting **6 potential rows** with data quality issues as a Proof of Concept for ING.
- Gained knowledge of credit risk metrics by collaborating daily with professionals and working in an international team from over 10 countries, which enriched my understanding of the **banking sector**.
- Developed **communication** and **delivery skills** by presenting several technical and business-level presentations, improving my ability to convey complex information effectively.
- Delivered the proof-of-concept within a tight schedule set by me and the project manager, where I learnt to set realistic **deadlines**.

- Applied **software development skills** through a summer internship as part of my Bachelor's degree, achieving a grade of 10/10.
- Gained a practical understanding of Android and MVVM Architecture in an **Agile Scrum Environment**, improving development efficiency and collaboration.
- Enhanced UI for **50%** of the key screens of their Android application, resulting in a more user-friendly experience and higher user satisfaction.
- Implemented Firebase Analytics, enabling better tracking and analysis of user behaviour.
- Developed **teamwork skills** by learning to use **Git** in a production environment, resulting in improved version control and collaboration.

REMARKABLE PROJECTS

Full Stack WebApp using Microservice Architecture

Enschede, Netherlands

Team Lead

Feb 2024 – Apr 2024

- Planned and developed a Full-Stack Web Application following **SOA microservices principles**, leveraging **Java** for the back end and **HTML, Thymeleaf, and JavaScript** for the front end, which enhanced the application's modularity and maintainability.
- Learnt and used **PostgreSQL** for database management, ensuring robust data storage and retrieval.
- Implemented containerization using **Docker** for each service, resulting in isolated and efficient service deployment.
- Integrated various APIs, including the **OpenAI API**, using GET/POST requests, enabling seamless third-party service interactions.
- Implemented synchronous and asynchronous service interactions, improving system **responsiveness**

Java Windows PC Game – Azul

Enschede, Netherlands

Team Lead

Apr 2023 – Jun 2023

- Collaborated in a team of 4 students to develop a desktop game in **Java**, applying **Test-Driven Development (TDD)** and Clean Code principles from start to finish.
- Implemented core gameplay logic and contributed to system architecture (MVC), a YAML CI/CD pipeline on GitLab, and applied successfully **4 design patterns**: Singleton, Observer, Proxy and Facade.
- Practised rigorous unit, integration and testing with JUnit, achieving **96.89% code coverage** and followed Agile iterations in **Jira** to plan, test, and refactor code.
- Focused on writing modular, readable, and maintainable code, emphasising quality over quick fixes assessed by the software quality platform **Sigrid**, with the score of **4.4/5**.

SKILLS

Programming: Java, Spring Boot, Python, Kotlin, SQL, Node.js

Technical: Backend Development, Firebase, Android, GIT, Machine Learning, Algorithms and Data Structures

Languages: English (C1 Cambridge Certified), Dutch (A1 level), Romanian (Mother tongue)

Soft Skills: Communicative, Problem Solving, Good Listener, Ambitious, Attention to Details, Critical Thinking.

Interests: Software/Web Development, React, AI, Tour Guiding, History, Calisthenics, Psychology

ACTIVITIES AND LEADERSHIP

Hackathons/Workshops: Won 2nd place in the IADE Creative Hackathon 2021, where we had to design and build in **one week** a Mobile application that helps Erasmus students integrate in Lisbon as Erasmus students.

Local Tour Guide: certified since 2021, improved communication skills while volunteering for the Cultural Capital Events in Timisoara, guiding tourists from 30+ countries around the world.

Sports: Captain of the football 7v7 team and the volleyball mix team from my University

Volunteering: Erasmus+ Projects regarding Sustainability (YouRoTrip 2022) and Alternative Learning (GeoFun 2022)